Graphs and Best Fit: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Slope from two points. Return →

The slope is $(9-3)/(4-1)=2$. The intercept is $3-2(1)=1$. The line is $y=2x+1$. **Check:** both points satisfy the equation.

Solution 2 — Linearization. Return →

Plot T^2 against L. The slope is $4\pi^2/g$, so $g=4\pi^2/\text{slope}$. **Check:** the intercept should be zero for an ideal small-amplitude simple pendulum; large amplitudes lengthen the period.

Solution 3 — Power law. Return →

The log-log slope is $n=2.5$. The intercept gives $a=10^{0.3}=2.0$. **Check:** the law is approximately $y=2x^{2.5}$.

Solution 4 — Least-squares slope. Return →

The points lie exactly on $y=2x+1$, so the least-squares slope is 2 and intercept 1. The fit is exact because the data have no scatter and are perfectly linear. **Check:** the residuals are zero.

Solution 5 — Residual. Return →

The predicted value is $2(3) + 1 = 7$. The residual is $8 - 7 = 1$, so the point is above the line. **Check:** residual = observed − predicted.

Solution 6 — R^2 caution. Return →

No. A curved residual pattern means the linear model misses structure. Try a physically motivated nonlinear model or add a term, and inspect the new residuals. **Check:** R^2 measures variance explained, not model correctness.

Solution 7 — Slope uncertainty. Return →

$\sigma_m = s/\sqrt{\sum (x_i - \bar{x})^2} = 0.2/\sqrt{10} = 0.063$. **Check:** a wider spread in x reduces slope uncertainty.

Solution 8 — Outlier effect. Return →

Least squares minimizes squared residuals, so a large residual has disproportionate influence. Investigate the point: check for transcription error, cloud, cosmic ray or a real physical effect. If it is excluded, state the rule and show the fit with and without it. **Check:** silent deletion is not responsible science.

Solution 9 — Exponential decay. Return →

The slope is $-\lambda$, so $\lambda = 0.1\ \text{yr}^{-1}$. The half-life is $\ln 2/\lambda = 6.93$ yr. **Check:** a negative slope on a ln-linear plot indicates decay.

Solution 10 — Physical intercept. Return →

A 0.2 cm intercept may indicate a ruler zero error, a systematic offset in the measurement procedure, or a real physical offset. Test by measuring a known zero length, reversing the ruler, or using an independent instrument. **Check:** the intercept is a diagnostic, not just a fitting nuisance.